Unitary Operators

Def.: $(V, \langle , \rangle)$ inner ~

V: $\mathbb{C}$-vector space

$T: V \to V$ is unitary (operator) $\iff \forall \underline{v}, \underline{w} \in V$,

$\langle T(\underline{v}), T(\underline{w}) \rangle = \langle \underline{v}, \underline{w} \rangle$

Remark: holds $\Rightarrow$ T preserves the norm of the vector $\underline{v} \in V$

i.e., $\forall \underline{v} \in V, \|T(\underline{v})\| = \|\underline{v}\|$

Prop.: T is unitary $\iff T^* \circ T = I$

Proof:

($\Rightarrow$)

Let $\forall \underline{x} \in V$,

then $\forall \underline{y} \in V$, $\langle (T^*T - I)(\underline{x}), \underline{y} \rangle$

$= \langle T^*T(\underline{x}), \underline{y} \rangle - \langle \underline{x}, \underline{y} \rangle$

$= \langle T(\underline{x}), T(\underline{y}) \rangle - \langle \underline{x}, \underline{y} \rangle$

$= 0$

nondegeneray of $\langle , \rangle$ $\Rightarrow (T^*T - I)(\underline{x}) = \underline{0}$, $\forall \underline{x} \in V$

$\Rightarrow T^*T - I$ is the zero operator

($\Leftarrow$) $\forall \underline{x} \in V$

$T^*T = I \Rightarrow T^*T(\underline{x}) = I(\underline{x}) = \underline{x}$

$\Rightarrow \forall \underline{y} \in V$, $\langle T^*T(\underline{x}), \underline{y} \rangle = \langle \underline{x}, \underline{y} \rangle$

$= \langle T(\underline{x}), T(\underline{y}) \rangle$

E.g.: $(V=\mathbb{C}^n, \langle\,,\rangle)$

Recall! $T: V \to V$, $T(\underline{x}) = A\underline{x}$

$\Rightarrow T^*: V \to V$, $T^*(\underline{x}) = \overline{A^t}\,\underline{x}$

Then T is unitary $\Leftrightarrow \overline{A^t} \cdot A = I_{n\times n}$

More explicity,

write $A = (\underline{u}_1 \mid \underline{u}_2 \mid \cdots \mid \underline{u}_n)$

then $\overline{A^t} \cdot A = I_{n\times n}$ implies:

$$\begin{pmatrix} \overline{\underline{u}_1^t} \\ \vdots \\ \overline{\underline{u}_n^t} \end{pmatrix} (\underline{u}_1 \mid \cdots \mid \underline{u}_n) = \begin{pmatrix} \langle \underline{u}_1, \underline{u}_1 \rangle & & \langle & \rangle \\ & \langle \underline{u}_i, \underline{u}_j \rangle & & \\ \langle & \rangle & \langle & \rangle \end{pmatrix}$$

Therefore,

$A = (\underline{u}_1 \mid \cdots \mid \underline{u}_n)$ ——

i.e., ——

Unitary Matrix

Def.: An $(n\times n)$ complex matrix A is unitary

$\Leftrightarrow \overline{A^t}A = I$

$\Leftrightarrow$ the columns of an orthogonal basis of $\mathbb{C}^n$

Lemma 1: A unitary operator $T: V \to V$
e.values $\lambda \in \mathbb{C}$ of T has $|\lambda| = 1$ ($|a+bi| = \sqrt{a^2+b^2}$

Proof:

Let $T(\underline{w}) = \lambda \underline{w}$ (eigen) (condition?)

Then $\langle T(\underline{w}), T(\underline{w}) \rangle = \langle \underline{w}, \underline{w} \rangle$

$\langle \lambda \underline{w}, \lambda \underline{w} \rangle = \cdots$

$\overline{\lambda} \lambda \langle \underline{w}, \underline{w} \rangle = \cdots$

$\langle \underline{w}, \underline{w} \rangle \neq 0 \Rightarrow \overline{\lambda} \lambda = 1$

(?) $\Rightarrow |\lambda|^2 = 1$

$\Rightarrow |\lambda| = 1$

(i.e., $\exists \theta \in \mathbb{R}, \lambda = e^{i\theta} = \cos\theta + i \sin\theta$)

Lemma 2: $T: V \to V$ is a unitary operator

$U \leq V$ is T-invariant

$\Downarrow$

$U \leq V$ is $T^{-1} = T^*$-invariant

Proof:

$T^*T = I \Rightarrow T$ is invertible and $T^{-1} = T^*$

① $\Rightarrow 0$ is not an e.value of T,

i.e., $\chi_T(0) \neq 0$

Write $\chi_T(x) = x^n + a_{n-1}x^{n-1} + \cdots + a_1 x + a_0$

Then $\Rightarrow a_0 \neq 0$

By C.-H., $\chi_T(T) = T^n + a_{n-1}T^{n-1} + \cdots + \sim + a_0 I = \underline{0}_{V \to V}$

$\Rightarrow I = -\frac{1}{a_0}T^n - \frac{a_{n-1}}{a_0}T^{n-1} - \cdots - \frac{a_1}{a_0}T$

$\Rightarrow I = \left(-\frac{1}{a_0}T^{n-1} - \cdots - \frac{a_1}{a_0}I\right)T$

$\Rightarrow T^{-1} = -\frac{1}{a_0}T^{n-1} - \cdots - \frac{a_1}{a_0}I$

$T^{-1} = T^* \Rightarrow T^* = -\frac{1}{a_0}T^{n-1} - \cdots - \frac{a_1}{a_0}I$

(WTS: $U \leq V$ is T^{-1}-invariant)

Take $\forall u \in U$, then $T^{-1}(u) =$ ~

(每项都 $\in U$)

$\Rightarrow T^{-1}(u) \in U$

$\Rightarrow$ ~

Lemma 3: $\begin{cases} T: V \to V \text{ is unitary} \\ U \leq V \text{ is } T\text{-invariant} \end{cases} \Rightarrow U^{\perp} \leq V$ is T-invariant

Proof:

Let $\forall \underline{w} \in U^{\perp}, \forall \underline{u} \in U$

$\langle \underline{u}, T(\underline{w}) \rangle = \langle T^*(\underline{u}), \underline{w} \rangle$

Lemma 2 $\Rightarrow T^*(\underline{u}) \in U \Rightarrow = 0$

$\Rightarrow T(\underline{w}) \in U^{\perp}$

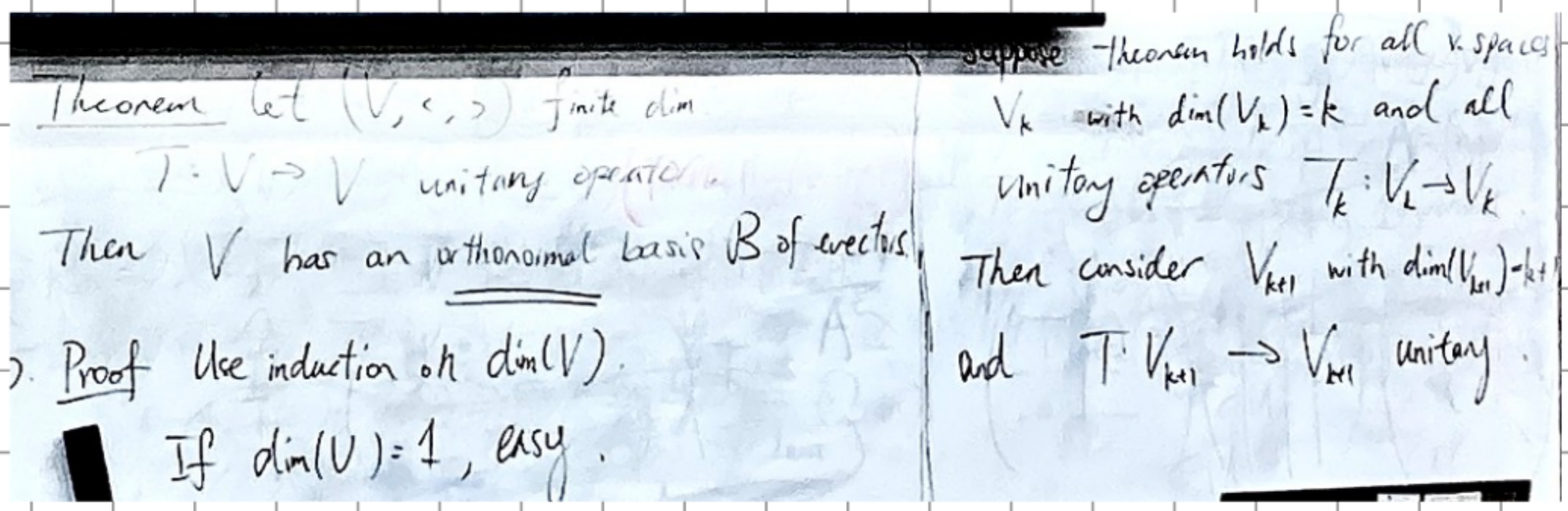
$\Rightarrow \sim$

Theorem 1: $(V, \langle , \rangle)$ is finite dimensional

$T: V \to V$ is a unitary operator

$\Downarrow$

V has an orthogonal basis B of e.vectors



Theorem Let $(V, \langle , \rangle)$ finite dim

$T: V \to V$ unitary operator

Then V has an orthonormal basis B of evectors

Proof Use induction on $\dim(V)$.

If $\dim(V) = 1$, easy.

Suppose Theorem holds for all k spaces V_k with $\dim(V_k) = k$ and all unitary operators $T_k: V_k \to V_k$.

Then consider V_{k+1} with $\dim(V_{k+1}) = k+1$ and $T: V_{k+1} \to V_{k+1}$ unitary.

Unitary Operators

Since V_{k+1} is a $\mathbb{C}$-v space, any operator $T: V_{k+1} \to V_{k+1}$ has an evector $y \in V_{k+1}$

Then let $U := \text{Span}\{y\}$ $(\dim(U) = 1)$ So we have

* $U^{\perp}$ has dimension k, since $\dim(U) + \dim(U^{\perp}) = \dim(V_{k+1})$
* $U^{\perp}$ is T-invariant by Lemma 3.

So $T|_{U^{\perp}}: U^{\perp} \to U^{\perp}$ satisfies

$\langle T|_{U^{\perp}}(\underline{w}), T|_{U^{\perp}}(\underline{w}') \rangle = \langle T(\underline{w}), T(\underline{w}') \rangle = \langle \underline{w}, \underline{w}' \rangle$

i.e. it is unitary.

Take $V_k = U^{\perp}$ and $T_k = T|_{U^{\perp}}$. By induction,

$\exists$ orthonormal basis $\{\underline{v}_2, \dots, \underline{v}_{k+1}\}$ of $U^{\perp}$

s.t. $T|_{U^\perp}(\underline{v}_i) = T(\underline{v}_i) = \lambda_i \underline{v}_i$

(i.e. $\underline{v}_i$ are evectors of T)

Then $\left\{ \frac{\underline{u}}{\|\underline{u}\|}, \underline{v}_2, \cdots, \underline{v}_{k+1} \right\}$

orthonormal basis of evectors of T in V.

By Thm. 1, we have $T_{BB} = \begin{pmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_{k+1} \end{pmatrix}$

Corollary: $A \in M_{n\times n}(\mathbb{C})$ is a unitary matrix

$\Downarrow$

$\exists$ unitary matrix P,

s.t. $P^{-1}AP = \begin{pmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_n \end{pmatrix}$, $\forall i$, $|\lambda_i| = 1$

Proof:

Let $(V = \mathbb{C}^n, \langle , \rangle)$, $\mathcal{C} = \{\underline{e}_1, \cdots, \underline{e}_n\}$ is usual basis of $V = \mathbb{C}^n$

and consider $T: V \to V$, $T(\underline{x}) = A\underline{x}$

Observations:

(i) $T_{\mathcal{C}\mathcal{C}} = A$

(ii) A is unitary matrix $\Rightarrow$ T is unitary operator

(iii) Thm. 1 $\Rightarrow$ $\exists$ orthogonal basis B consisting of e.vectors of T,

i.e. $T_{BB} = \begin{pmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_n \end{pmatrix}$

Note that $T_{BB} = \mathcal{C}_{BB}^{-1} T_{\mathcal{C}\mathcal{C}} \mathcal{C}_{BB}$

$\mathcal{C}_{BB} = (\underline{b}_1 | \cdots | \underline{b}_n)$

(接上)

Therefore, $\begin{pmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_n \end{pmatrix} = \varphi_{BB}^{-1} A \varphi_{BB}$ where ① φ_{BB} is a unitary matrix